UCF Horse Plinko Cyber Challenge

Competition Write-Up • October 5, 2024

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Overview

On October 5th, 2024, I competed in the UCF Horse Plinko Cyber Challenge as part of a three-person team — one teammate was unable to attend, leaving us a player short for the entire event. Despite this disadvantage, our team placed 13th out of 22 competing teams with a final score of 1,816.40 points, a result I am proud of given the circumstances.

This was a Collegiate Cyber Defense Competition (CCDC)-style event in which teams were simultaneously required to maintain system uptime, respond to adversarial attacks from a live red team, and complete timed challenge tasks called injections. The environment demanded strong multitasking, real-time decision-making, and effective teamwork under pressure.

Technical Skills Demonstrated

Throughout the competition, I gained hands-on experience with a wide range of tools, technologies, and security concepts, including:

- Remote Desktop Protocol (RDP) and Remote Desktop Connection for managing Windows-based systems
- Windows Server Manager and Windows Security Center for monitoring system health and configuring defenses
- Computer Management for user account administration, service control, and process monitoring
- Linux command-line administration, including file permissions, process management, and service configuration
- Network protocols including FTP, and understanding their attack surfaces in a live adversarial setting
- Firewall configuration and real-time rule updates to block malicious outbound connections based on active log analysis
- VNC (Virtual Network Computing) — identified post-event as the mechanism used by the red team to remotely observe and control our systems
- C2 (Command and Control) server concepts, including how attackers establish persistent access and exfiltrate data
- MySQL database administration and securing database services against unauthorized access

- Log analysis and rapid troubleshooting using online resources and error code lookups

This competition also strengthened my soft skills, particularly in cross-functional communication, task delegation, and adaptable planning within a time-constrained team environment.

Team Strategy & Execution

Initial Setup

At the start of the event, our first priority was credential hygiene. We immediately changed all default passwords across every system, assigning unique passwords per machine and storing them securely in the private Discord channel provided by the event organizers. This was a foundational hardening step intended to limit the red team's ability to use default or previously known credentials.

Defensive Operations

Our ongoing defensive approach included several key practices:

- Reactive firewall rule updates: Whenever a suspicious or malicious IP was identified in system logs, we immediately created outbound block rules to cut off the connection.
- Process monitoring: We regularly inspected running processes on each system to identify and terminate any unauthorized or suspicious activity.
- User account auditing: We continuously monitored user accounts for unauthorized additions, re-enabled disabled accounts, or privilege escalation. When suspicious accounts were found, we removed them, deleted associated files, revoked elevated permissions, disabled the accounts, and rotated credentials to prevent re-entry.
- Application integrity checks: We monitored installed applications for any unauthorized software deployed by the red team during the event.

Task Allocation & Adaptability

Our team initially split responsibilities with two members focused on injection challenges to maximize point gains, while the third member handled system hardening and active defense. As the red team's attacks increased in frequency and complexity over the course of the event, I transitioned from injections to defensive operations, shifting us to a one-injection, two-defense structure.

This allocation remained fluid throughout the event. If defensive pressure eased or a teammate hit a roadblock on an injection, we dynamically reallocated to maximize progress. This adaptability allowed us to remain competitive despite operating with only three team members.

Challenges & Lessons Learned

VNC Exploitation

The most significant challenge we faced was the red team's use of VNC software installed covertly on our systems. Because none of us had prior exposure to VNC in a red-versus-blue context, we were unaware that the red team had a live view of our screens throughout much of the event. This allowed them to observe our password changes in real time, identify our defensive strategies as we implemented them, and in some cases remotely undo our countermeasures.

This was our most impactful knowledge gap during the event. After the competition concluded, the red team walked us through exactly how VNC was deployed, how they used it, and how we could have detected and removed it. This became one of the most valuable learning experiences of the entire event and is a gap I have since prioritized filling.

Antivirus Installation

Our team did not install antivirus software during the competition, as we were uncertain whether doing so was permitted under the competition rules and did not want to risk disqualification. After the event, we learned that installing security tools such as Malwarebytes was entirely allowed. Had we known, deploying antivirus would have significantly improved our ability to detect and remove malicious software that the red team had introduced to our systems.

Key Takeaways

This competition was my first hands-on experience in a live red-versus-blue cyber defense environment, and it was an invaluable learning opportunity. The key lessons I am taking forward include:

- Always verify the rules regarding tool installation before and during competition — don't assume restrictions that aren't there.
- VNC and remote access tools are a critical attack vector in blue team defense scenarios and should be one of the first things checked during initial system triage.
- Effective communication and flexible task allocation within a team are just as important as individual technical skill in a time-pressured environment.
- Log analysis and real-time firewall management are core blue team competencies that I was able to practice under realistic conditions.

Despite finishing 13th, our team operated with only 75% of our roster and still placed in the top 60% of the field. The experience directly reinforced skills relevant to roles in

security operations, incident response, and blue team defense — and highlighted specific areas such as endpoint monitoring and remote access threat detection that I have since prioritized in my ongoing studies.

Photo credited to Michael & Betty Freeland Photography

This is a photo of everyone who participated. I am circled in red.